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Design and circuit model analysis of load coil embedded resonant inductively coupled wireless power transfer system

Pragyan P. Mohanty, Sushree S. Biswal✉, Durga P. Kar & Satyanarayan Bhuyan

Magnetic Resonance Coupled (MRC) based conventional two-coil Wireless Power Transfer (WPT) systems are fraught with difficulties due to restricted power transfer distance and inadequate strategic control over power flow to the load. This is because an increase in air-gap causes a substantial fall in the coupling coefficient between the coils, which in turn reflects a large decline in the delivered load power and power transfer efficiency (PTE). In order to overcome the stated issues, a unique WPT approach that integrates three coil structures with an additional load coil with fewer turns is proposed and the justification as an alternative viable solution has been outlined in this research. A meticulous load coil comprised three-coil WPT circuit model investigation with in-depth mathematical analysis and the key performance characteristics has been substantiated through experimental measurements. The systematic equivalent circuit-model analysis of two-coil and three-coil WPT systems unveil the inclusive link/impacts of functional parameters like mutual coupling, driving frequency as well as resistive load on the output load power and efficiency of WPT system. The power delivered to load (PDL) through this explored technique is observed considerably higher i.e ~16 Watt as opposed to ~4.5 Watt in conventional WPT system, hence appealing higher power transfer range. The corroborated experimental and simulation results with proper illustration will be helpful in providing design strategies for forthcoming MRC based WPT system with maximum power delivery ability.

Keywords Circuit model analysis, Load coil embedded receiver, Mutual coupling, Magnetic resonance coupling, Wireless power transfer

Magnetic coupling-based wireless power transfer has gained popularity as a study topic in recent decades due to its versatility and ability to function in all weather conditions. This technique is becoming more and more common since it is well suited for a variety of charging applications with different power needs[1–5]. Due to its comprehensible nature, the favoured WPT technology is used for micro robots, integrated circuit power supplies, medical implants, electronic appliances, cell phones, laptops, underwater equipment's and road-way electric vehicles[6–10].In inductive method, the air gap between the transmitting and receiving coils causes visible effects such as weak mutual inductance, large leakage flux, radiation loss, sluggish charging and heating effects[11–13].To overcome the difficulties, an alternate method in term of resonant inductive coupling WPT schemeis developed in order to satisfy customer needs with the least amount of compromise[7,14–16]. Unlike the conventional inductive type, resonant WPT allows more efficient power transfer over a greater distance with little to no radiation from nearby off-resonant objects. Similar to conventional two-coil method, WPT system comprising of four-coils are found more appropriate in certain situations[17–21]. The exceptionality of this configuration is in providing isolation between source and enabling power transfer over larger range by using additional coils. If the load coil comprised 3-coil WPT system is compared with the conventional WPT system, it can be apprehended that the three-coil configuration overcomes the challenges of the two-coil system and includes the benefits of the four-coil system. The inclusion of a load coil in the three-coil WPT architecture enables controlled and flexible power delivery to the load, as the position of the load coil can be adjusted independently while maintaining fixed transmitting and receiving coil locations. In order to determine the best WPT methodology and chosen technologies based on their applicability, this calls for the pursuit of methodical research.

Faculty of Engineering and Technology, ITER, Siksha 'O' Anusandhan Deemed to be University, Bhubaneswar 751030, India. ✉email: sushreesangitabiswal@soa.ac.in

This article delineatesdeeper analytical content with in-depth circuit model analysis endorsing the purported advantages of the proposed three coil WPT architecture over the traditional two-coil, and the key performance characteristics have been substantiated through experimental measurements, supporting the technical rigor of the study.It provides the inclusive link between the functional parameters like mutual coupling, operating frequency and resistive load and the resulting impact on the WPT systems' performance like output load power and PTE through the systematic equivalent circuit model analysis of two- and three-coil WPT system. Furthermore, this work clarifies the impact of receiver coil and load coil position adjustments on PDL and PTE as well as the enhancement of flexibility and controllability over power flow in load coil embedded three-coil structures. To confirm these results, an experimental three-coil WPT system setup was built on a test bench. The experimental results based on this configuration then confirmed the theoretical conclusions. The study concludes by highlighting the benefits of increasing the efficiency of resonant WPT systems by employing three coils to power many electrical devices at once.

The suggested three-coil WPT method is found appropriate for wireless transmission after a thorough comparison with the conventional one. Furthermore, the investigated 3-coil WPT system found delivering larger load power as compared to 2-coil structure even with extended power transfer distance under similar operating conditions. The decrease in power transfer efficiency observed in 3-coil WPT system is primarily arises from the increased effective transfer distance between the transmitting and load coil as compared to traditional 2-coil WPT system. Consequently, the need for repeated redesign or reconfiguration of the coil geometry to achieve controlled power transfer under varying load conditions which is a common limitation in conventional two-coil WPT system can be effectively mitigated.

This study is organized as follows: Section "Detailed theoretical analysis" presents the detailed theoretical analysis of proposed three-coil WPT system. Section "Experimental design setup" discusses the experimental setup. Section "Results and discussion" provides a detailed description of the analytical modelling results that are corroborated by the experimental results. In Section "Conclusions", appropriate conclusions and future directions are outlined.

Detailed theoretical analysis

In order to disclose the characteristics of proposed three-coil and conventional two-coil WPT systems, the transmission efficiency (PTE) of both the systems have been deduced by utilizing multi-coil WPT system architecture. An impedance matrix is used to express the relationship between the multi-coil WPT system's series *RLC* resonators and the coupling between distant coils (the transmitting and load coils) are taken into consideration while calculating the transmission efficiency[20]. For the purpose of comparison, the critical coupling condition that determines higher transmission efficiency in two- and three-coil WPT systems as well as the requirements for a three-coil WPT system to have higher transmission efficiency are clarified. The circuit equivalent model of the multi-coil WPT with n resonators is depicted as shown in Fig. 1. V_S is the source voltage and each resonator includes a coil L, a capacitor C and a resistor R_p(parasitic resistor). $R_S$and R_L are the source and load resistance. All the coils are mutually coupled with each other(M_{ij}).

Applying KVL, the mathematic model of the n-coil WPT system in matrix form is expressed as$ZI = V$.

$$\begin{bmatrix} R_1 + j\omega L_1 + \frac{1}{j\omega C_1} & -j\omega M_{12} & \cdots & -j\omega M_{1n} \\ -j\omega M_{21} & R_2 + j\omega L_2 + \frac{1}{j\omega C_2} & \cdots & -j\omega M_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -j\omega M_{n1} & -j\omega M_{n2} & \cdots & R_n + j\omega L_n + \frac{1}{j\omega C_n} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} V_S \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (1)$$

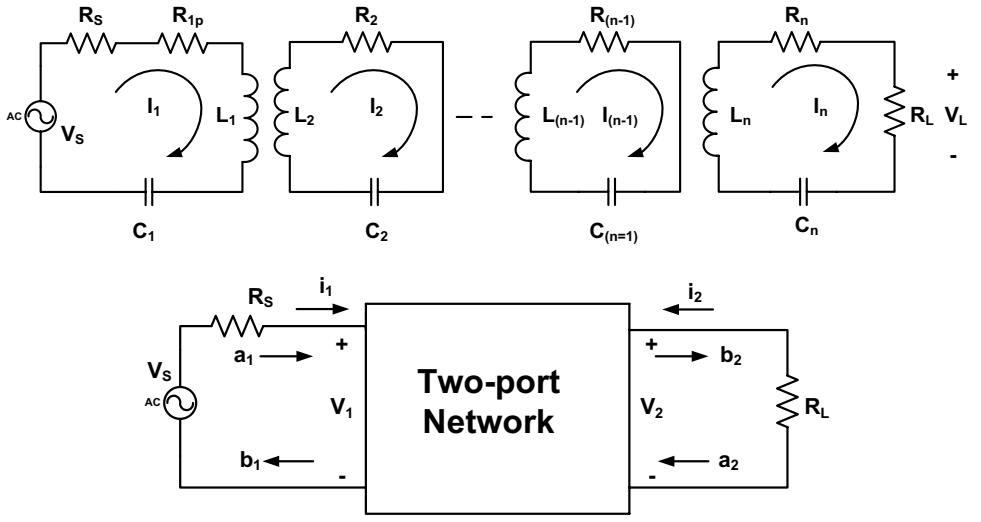


Fig. 1. Equivalent circuit model of multi-coil WPT system and its two-port configuration.

Where, the impedance matrix Z is of the order n by n and total resistance seen at the input of each individual resonator is represented by R_i and calculated as $R_i = R_{ip}$ except $R_1 = Rs + R_{1p}$ and $R_n = R_n + R_L$. Moreover, the n-coil WPT system can be modelled equivalently as combination of a generator, a load and a two-port network as shown in Fig. 1. The PTE of the n-coil WPT system is calculated by[22]

$$\eta = |S_{21}|^2 = \left|\frac{b_2}{a_1}\right|^2_{a_2=0} \tag{2}$$

Where,

$$\begin{cases} a_1 = \dfrac{v_1 + R_S i_1}{2\sqrt{R_S}} \\ b_2 = \dfrac{v_2 + R_L i_2}{2\sqrt{R_L}} \end{cases} \tag{3}$$

Hence, the PTE of (2) becomes

$$\eta = |S_{21}|^2 = 4R_S R_L \left|\frac{I_n}{V_S}\right|^2 = 4R_S R_L \left|\left[Z^{-1}\right]_{n1}\right|^2 = 4R_S R_L \left|\frac{Z_{n1}}{\Delta}\right|^2 \tag{4}$$

The ratio I_n/V_S can be calculated from the nth row and the first column element of the inverse of the impedance matrix, Z_1. Z_{n1} is the adjugate matrix and Δ is the determinant of the matrix. By using this method, the complete inverse of the impedance matrix is not needed rather only n^{th}row and the first column element of the inverse impedance matrix are essential. Hence, PTE calculation with higher order impedance matrix of the multi-coil WPT system because of the number of coils still can be easily calculated.The impedance matrix for the two-coil WPT system with series RLC structure resonator is given by

$$Z = \begin{bmatrix} R_1 + j\omega L_1 + \frac{1}{j\omega C_1} & -j\omega M_{12} \\ -j\omega M_{21} & R_2 + j\omega L_2 + \frac{1}{j\omega C_2} \end{bmatrix} \tag{5}$$

The modified impedance matrix when each resonator resonates at the operating frequency is as follows:

$$Z = \begin{bmatrix} R_1 & -j\omega M_{12} \\ -j\omega M_{21} & R_2 \end{bmatrix} \tag{6}$$

Following (4), the PTE of the two-coil WPT system is calculated as

$$\eta = |S_{21}|^2 = 4R_S R_L \left|\frac{j\omega M_{12}}{R_1 R_2 + \omega^2 M_{12}^2}\right|^2 \tag{7}$$

The mutual inductance and the loaded quality factor of transmitting and receiving resonators are respectively expressed as follows:

$$\begin{cases} M_{12} = k_{12}\sqrt{L_1 L_2} \\ Q_1 = \dfrac{\omega_0 L_1}{R_{1p} + R_S} = \dfrac{\omega_0 L}{R_1} \\ Q_2 = \dfrac{\omega_0 L_2}{R_{2p} + R_L} = \dfrac{\omega_0 L}{R_2} \end{cases} \tag{8}$$

Substituting (8) in in (7), the PTE is obtained by

$$\eta_{2-coil} = \frac{4R_S R_L}{R_1 R_2} \frac{k_{12}^2 Q_1 Q_2}{\left(1 + k_{12}^2 Q_1 Q_2\right)^2} \tag{9}$$

Replacing $k^2_{12}Q_1Q_2$ with A [which is usually referred as the figure-of-merit (*FOM*) and has always a positive value ($A \geq 1$)], the PTE of the two-coil WPT system can be written as:

$$\eta_{2-coil} = \frac{4R_S R_L}{R_1 R_2} \frac{A}{(1 + A)^2} \tag{10}$$

Differentiating with respect to A, the PTE is found to be maximum at $A = 1$, hence, the critical coupling, $k_{critical} = 1/\sqrt{Q_1 Q_2}$.

Similarly, the PTE of the three-coil WPT system can be derived as

$$\eta_{3-coil} = |S_{21}|^2 = 4R_S R_L \left|\frac{\omega^4 M_{12}^2 M_{23}^2 + \omega^2 M_{13}^2 R_2^2}{\Delta}\right|^2 \tag{11}$$

Where, Δ is the determinant of the impedance matrix of the 3-coil WPT system, and is expressed as

$$\Delta = \left(R_1R_2R_3 + \omega^2M_{23}^2R_1 + \omega^2M_{12}^2R_3 + \omega^2M_{13}^2R_2\right)^2 + 4\omega^6M_{12}^2M_{23}^2M_{13}^2 \tag{12}$$

The PTE as given in (11) can be rewritten by using loaded Q of the resonators and the coupling coefficient

$$\eta_{3-coil} = \frac{4R_SR_L}{R_1R_2}\frac{k_{12}^2k_{23}^2Q_1Q_2^2Q_3 + k_{13}^2Q_1Q_3}{\left(1 + k_{12}^2Q_1Q_2 + k_{13}^2Q_1Q_3 + k_{23}^2Q_2Q_3\right)^2 + 4k_{12}^2k_{23}^2k_{13}^2Q_1^2Q_2^2Q_3^2} \tag{13}$$

The expression in (13) can be rewritten using $A = k^2_{12}Q_1Q_2$, $B = k^2_{23}Q_2Q_3$ and $C = k^2_{13}Q_1Q_3$as the *FOM* between the resonators (first and second, second and third and first and third) respectivelyas follows:

$$\eta_{3-coil} = \frac{4R_SR_L}{R_1R_3}\frac{AB + C}{(1 + A + B + C)^2 + 4ABC} \tag{14}$$

For PTE as shown in (14) to be maximum, $\partial\eta/\partial A = \partial\eta/\partial B = 0$, which results the following output,

$$\begin{cases} \dfrac{\partial\eta}{\partial A} = (1 + A + B + C)\left(B^2 + B + BC - AB - 2C\right) - 4BC^2 = 0 \\ \dfrac{\partial\eta}{\partial B} = (1 + A + B + C)\left(A^2 + A + AC - AB - 2C\right) - 4AC^2 = 0 \end{cases} \tag{15}$$

Comparing both the expression, it can be implied that $A = B$ and the symmetric three-coil WPT delivers comparatively higher PTE than asymmetric system. In case of symmetric structure, the PTE can be written as

$$\eta_{3-coil} = \frac{4R_SR_L}{R_1R_3}\frac{A^2 + C}{(1 + 2A + C)^2 + 4A^2C} \tag{16}$$

It can also be noted that equation (16) becomes the PTE of a two-coil WPT system if $A = B = 0$. Moreover, comparison between PTE of two-coil and three-coil discloses the suitability of individual WPT system based on the application.

To find out the situation where three-coil WPT system is preferred over two-coil WPT system, let us assume

$$\begin{aligned} &\eta_{2-coil} < \eta_{3-coil} \\ &\Rightarrow \frac{4R_SR_L}{R_1R_3}\frac{C}{(1 + C)^2} = \frac{4R_SR_L}{R_1R_3}\frac{A^2 + C}{(1 + 2A + C)^2 + 4A^2C} \end{aligned} \tag{17}$$

Solving for C gives the condition which ensure 3-coil WPT system facilitating higher PTE than its counter-part two-coil WPT system.

$$0 < C < \frac{A}{4 + 3A} \tag{18}$$

The parameter C is the *FOM* between transmitting and load resonator, which particularly specifies the transmission distance of the three-coil WPT system. Lower the C value means higher the transmission distance and vice versa. Moreover, equation (18) reveals the impression that the three-coil WPT system facilitates higher PTE than its counterpart two-coil system when the transmission distance is more and two-coil WPT system is best suited for a shorter transmission distance as expected. Furthermore, term A outline the critical limit at which both two- and three-coil WPT systems have equal PTE.

Case-I

If $A >> 1$, upper limit of C reaches *1/3* and equation (18) reduces to $0 < C < \frac{1}{3}$,or $0 < k_{13} < \frac{1}{\sqrt{3Q_1Q_3}}$ and the three-coil WPT system can be used at shorter transmission range. If $k_{13} > k_{critical}/\sqrt{3}$, a two-coil WPT system is recommended otherwise a three-coil WPT system suitable. Hence, $k_{critical}/\sqrt{3}$ is the critical condition for deciding the appropriate WPT system to use.

Case-II

If the receiving coil is not coupled with the load coil ($k_{23} = 0$ or $B = 0$), then the PTE as shown in equation (14) becomes

$$\eta_{3-coil} = \frac{4R_SR_L}{R_1R_3}\frac{C}{(1 + A + C)^2} \tag{19}$$

The term A in the denominator reduces the PTE. When all the resonators of the three-coil WPT are tuned to same frequency and coupling k_{12} exist, then the PTE is inferior to two-coil WPT system.

Case-III

If $A = 0$ and $C \approx 0$, the three-coil WPT system still provide higher PTE even with weak coupling between transmitting and load coil. This is due to the fact that a little bit of power directly transferred from the transmitting to load coil and most of the power is transferred to the load through the receiving coil.

Case-IV

When all the resonators are weakly coupled with each other ($A \approx 0$ and $C \approx 0$), it results a very low PTE.

The proposed load coil comprised three-coil WPT system is a special case of multi-coil WPT system in which the non-resonated load coil carries fewer coil turns as compared to transmitting and receiving coil.In contrast to prior literature[20], the primary contribution of this research lies in the analytical derivation of delivered load power and PTE for a multi-coil WPT system in which the load coil is explicitly modelled as a non-resonant coil, rather than as part of a fully resonant network as commonly assumed. The equivalent circuit model of two-coil and load coil based three-coil WPT system is further analysedto outline the inclusive link between the functional parameters like mutual coupling, operating frequency and resistive load and the resulting impact on the WPT systems' performance like output load power and PTEas well as to ensure both theoretical rigor and effective system-level utility in perception to multi-receiver WPT system.

Equivalent circuit model of two-coil resonant WPT system

The equivalent circuit model of the conventional two-coil MRC-WPT system is depicted in Fig. 2. R_s, R_L, and (R_t, R_r) are the source resistance, load resistance and effective resistance of transmitting and receiving coil respectively. V_s represent the *RMS* value of the source voltage. Both, transmitting coil and the receiving coil are of same dimension and are compensated through the external capacitors Ct and Cr correspondingly. With mutual coupling between the coils (M_{12}), the coupled circuit is resonated at a frequency (f_r).

Based on the KVL, the two-coil MRC-WPT system is established by the following matrix

$$\begin{bmatrix} R_s + R_t + j\omega L_t + {}^{1}/_{j\omega C_t} & -j\omega M_{12} \\ -j\omega M_{tr} & R_r + j\omega L_r + {}^{1}/_{j\omega C_r} + R_l \end{bmatrix} \begin{bmatrix} I_t \\ I_r \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix} \tag{20}$$

$$\begin{cases} I_t = \dfrac{V_s\,(R_r + R_l)}{(R_s + R_t)\,(R_r + R_l) + \omega^2 M_{tr}^2} \\ I_r = \dfrac{j\omega M_{tr}\left[\frac{V_s(R_r+R_l)}{(R_s+R_t)(R_r+R_l)+\omega^2 M_{tr}^2}\right]}{(R_r + R_l)} \end{cases} \tag{21}$$

$$\begin{cases} P_{in} = |I_t|^2 R_t + |I_r|^2 R_l \\ P_{out} = |I_r|^2 R_l \\ \eta = \dfrac{P_{out}}{P_{in}} \times 100 \end{cases} \tag{22}$$

Equivalent circuit model of three-coil resonant WPT system

The circuit configuration of MRC based three-coil WPT system is illustrated in Fig. 3. The three-coil arrangement includes an extra load coil instead of the single receiving coil found in conventional designs.Moreover, in comparison to both transmitter and receiver coil, the extra coil is usually designed with fewer turns.With prime objective of expeditingcontrol and flexibility in transferring power flow to the load, additional benefit in term of efficiency is also promoted by making it easier to vary the air-gap between the receiving and load coil (M_{rl}).

$$\begin{bmatrix} R_0 + j\omega L_t + {}^{1}/_{j\omega C_t} & -j\omega M_{tr} & 0 \\ -j\omega M_{tr} & R_0 + j\omega L_r + {}^{1}/_{j\omega C_r} & -j\omega M_{rl} \\ 0 & -j\omega M_{rl} & j\omega L_l + R_l \end{bmatrix} \begin{bmatrix} I_t \\ I_r \\ I_l \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \end{bmatrix} \tag{23}$$

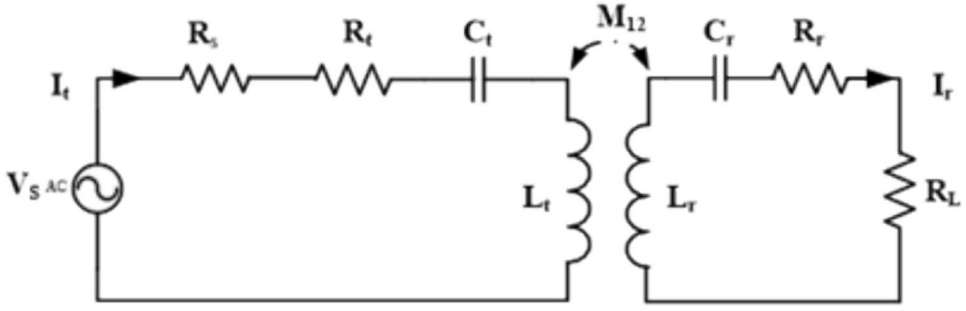


Fig. 2. Equivalent circuit model of two-coil WPT system.

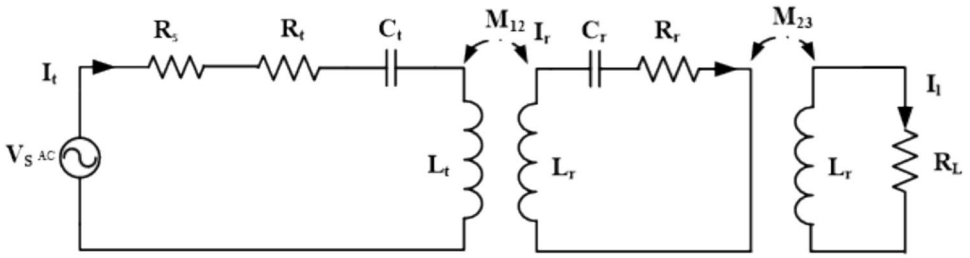


Fig. 3. Equivalent circuit model of three-coil WPT system.

The solution of the above equation is as follow:

$$\begin{cases} I_t = \dfrac{\left(R_0 + j\omega L_r + {}^{1}/_{j\omega C_r} + \omega^2 M_{rl}^2\right) V_s}{X + \omega^2 M_{tr}^2 R_l} \\ I_r = \dfrac{-j\omega M_{tr} R_l V_s}{X + \omega^2 M_{tr}^2 R_l} \\ I_l = \dfrac{\omega^2 M_{rl} M_{tr} V_s}{X + \omega^2 M_{tr}^2 R_l} \end{cases} \tag{24}$$

Where

$$X = \left(R_0 + j\omega L_t + {}^{1}/_{j\omega C_t}\right)\left(R_0 + j\omega L_r + {}^{1}/_{j\omega C_r} + \omega^2 M_{rl}^2\right)$$

$$\begin{cases} a = j\omega L_t + {}^{1}/_{j\omega C_t} \\ b = j\omega L_r + {}^{1}/_{j\omega C_r} \end{cases}$$

With the load coil of few turns as discussed above, the inductive reactance is very less and can be neglected as compared to load and hence will be considered as a resistive load:

$$j\omega L_l + R_l = R_l \tag{25}$$

The injected input power to the transmitter coil and the outputload power are calculated as follows:

$$\begin{cases} P_{in} = |I_t|^2 R_0 + |I_r|^2 R_0 + |I_l|^2 R_l \\ \quad = \dfrac{\left[\left(R_0 + b + \omega^2 M_{rl}^2\right)^2 V_s^2\right] R_0 + \omega^2 M_{tr}^2 R_l^2 V_s^2 R_0 + \omega^4 M_{rl}^2 M_{tr}^2 V_s^2 R_l}{\left(X + \omega^2 M_{tr}^2 R_l\right)^2} \\ P_{out} = |I_l|^2 R_l \end{cases} \tag{26}$$

The overall power transfer efficiency is calculated as:

$$\eta_{one-load} = \frac{P_{out(one-load)}}{P_{in(one-load)}} = \frac{|I_l|^2 R_l}{|I_t|^2 R_0 + |I_r|^2 R_0 + |I_l|^2 R_l} \times 100 \tag{27}$$

From the above equation (27), the load resistance which ensure maximum power transfer efficiency can be obtained through, $\partial\eta / \partial R_l = 0$ and the optimal load resistance is:

$$R_{l-optimal} = \frac{(\omega M_{rl})^2}{\sqrt{R_0^2 + (\omega M_{tr})^2}} \tag{28}$$

From the results obtained as shown in equations, (26), and (27) and (28), the design parameters like M_{tr}, M_{rl}, R_l and frequency of operation (ω) are significantly influencing the performance of the WPT system and hence needs to be considered carefully. This is worth to note that random considerations of these parameters not only impact the overall performance but also the design and analysis accuracy of the WPT system. Moreover, the load value required to be evaluated precisely considering the optimal load condition as obtained in equation (26) and (28). Moreover, these findings are critical and have a noteworthy impact in designing a suitable WPT system.

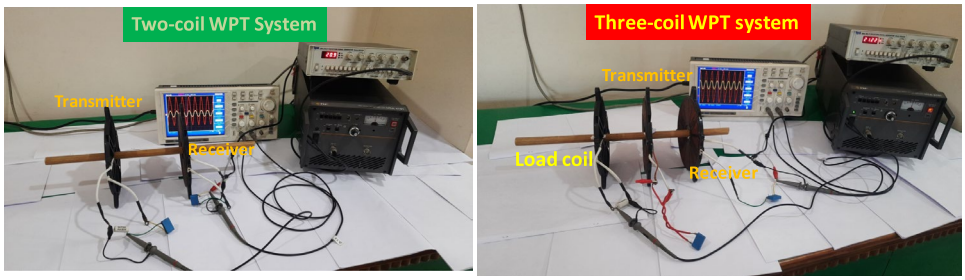


Fig. 4. (**a**)-(**b**). Experimental setup of 2-coil and load coil involved 3-coil WPTS.

Functional parameters	Values
Source voltage (V_s)	*7.1 V (R.M.S)*
Resonant frequency (f_r)	*20.2 kHz*
Transmitter and receiver coil's inductance (L_t &L_r)	*168 μH*
Compensating capacitor at transmitter and receiver coil (C_t&C_r)	*377 nF*
Effective series resistance of transmitter and receiver coil (R_t&R_r)	*0.135 Ω*

Table 1. Circuit model parameters of WPT system.

Experimental design setup

The experimental design setup for both two- and three-coil WPT systems are demonstrated in Fig. 4 (a-b).In the experimental prototype the wireless link comprises of transmitting coil, receiving coil and the load coils. Thecircular spiral transmitting and receiving coils aremade up of litz wire (36 AWG) and fixed in design with similar dimensions. With 28 number of turns each, the inner and outer radius of these coils are 2 cm and 10 cm respectively, whereas the load coil is similarly designed with 5 turns only of inductance 5.68 μH with inner and outer radius of 8 cm and 10 cm respectively.

The quality factors of transmitting and receiving coils are same ($Q_t = Q_r = 164.02$). The inductance value of the load coil (L_l) may vary in different cases by changing its size and turn number in order to achieve the desired mutual inductance between receiving coil and the load coil (M_{rl}). The function generator acts as the source provides the necessary radio frequency signal which further amplified by the high frequency power amplifier. The output of the high frequency power amplifier is connected to the transmitter coil through the compensating capacitor. This is accomplished through the use of a compensation network, which connects an additional capacitor to guarantee tuning at their resonance frequency and permits maximum amount of exciting current to flow through the circuit. The receiver coil is usually tuned to the transmitter coil so as to deliver power to the resistive load connected across it. The transmitter and receiver coils considered for this experimental setup are similar in dimension. The parameters used for the experimental results corresponding to the equivalent circuit model of the WPT system are listed as shown in Table 1.

Results and discussion

The schematic representation of the resonant WPT system as shown in Fig. 1 and its practical setup in Fig. 4 (a-b) showcasing the correlation of circuit model outcomes with experimental measurements in this section. A thorough grasp of the operation and underlying mechanisms of the WPT system is provided as per the theoretical and experimental analysis.

Using the analytical results based on the circuit models, the performance characteristics of WPT system comprising load coil have been evaluated and compared with the conventional two-coil system. Figure 5 and 6 shows the frequency characteristic of delivered load power for both two- and three-coil WPT systems, respectively, for different load situations.

The model findings are taken into consideration at a fixed mutual inductance value of (M_{12}= M_{23}=*12.25μH*) for both two- and suggested three-coil WPT system. Under similar operating resonant frequency of *20.2 kHz*, the power across the load in three-coil WPT system is found comparatively lower for higher load values than lower range.In addition, keeping other parameters alike, the frequency splitting phenomenon,which is clearly observed for higher load in three coil system witnessed at smaller electric load in traditional two-coil WPT system. Optimal load condition for maximum output power is also perceived in both systems. Keeping resonant frequency and electric load value for both cases same at *20.2 kHz* and *2.2 Ω*, suggested three-coil WPT system delivers an optimum load power of *15.82 W* in contrast to *4.22 W* for conventional method and any deviation from the ideal load value the output power drastically deteriorates. From the results, it can be asserted that the recommended 3-coil structure enables higher power transmission wirelessly. Fig. 7

The usefulness of the employed model is further supported by the good agreement between the simulated data and the experimental results. For example, when the resonant frequency is *20.2 kHz* and the load is *2.2 Ω*, the delivered power measures *15.98 W*, which is in good agreement with the data that was conjectured. The

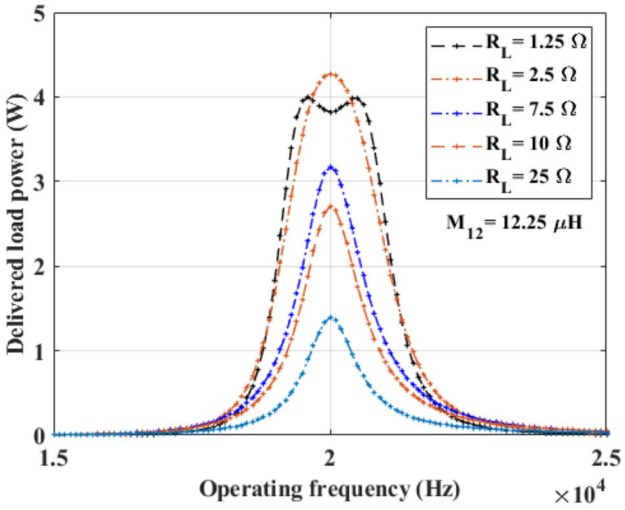


Fig. 5. PDL versus frequency in conventional WPT system.

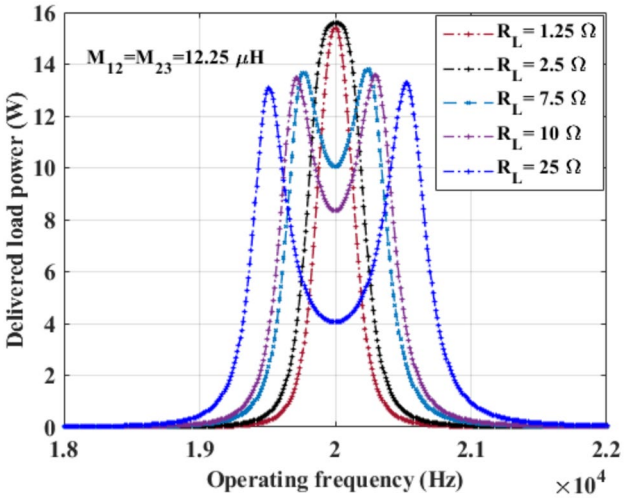


Fig. 6. PDL versus frequency in load coil involved (three-coil) WPT system.

results show that the delivered load power improves, when a load coil is added. It is clear from this comparison that there are real benefits to incorporating a load coil into the WPT system's architecture.

The overall PTE of both two- and three-coil WPT systems evaluated at a fixed mutual inductance of *12.25 μH* corresponds to operating frequency with varying electric load are elucidated in Fig. 8 & 9. The efficiency of the suggested three-coil WPT system is witnessed significantly lower as compared to the conventional method. The PTE of both systems progressively drops as load deviates from its ideal value. Fig. 10 displays the results of the experimental measurement that was carried out under the same operating parameters. As illustrated, the PTE of two-coil is measured as *86.4%* whereas *80%* in case of three-coil WPT system, when both the systems are operated at their resonant frequency and load of *2.2 Ω*.

In order to corroborate the findings from frequency characteristics of PDL and PTE as discussed earlier, the load power of both the models operated at their tuned frequency of *20 kHz* are mapped to different mutual coupling for varied load values. The same are depicted in Fig. 11 & Fig. 12 respectively. The findings as presented in the results clearly show that design parameter mutual coupling, a measure of the transmission range of wireless link indirectly determines the ideal load resistance. The ideal load values progressively rise with decrease in power transfer range. Remarkably, power transmission over a larger range necessitates a lesser electric load and shorter demands a bigger electric load to deliver optimum load power. The experimental observations shown in Fig. 13 supports the observation that the power delivery capability of three-coil is significantly higher compared to the traditional method of wireless power transmission (two-coil).

It is apparent that the three-coil method allows for higher power transmission, but the ideal load condition corresponding to optimum PTE varies. Figs. 14 and 15 show the PTE of both two- and three-coil WPT systems operated at their tuned frequency of 20.2 kHz under various load situations correspond to various mutual

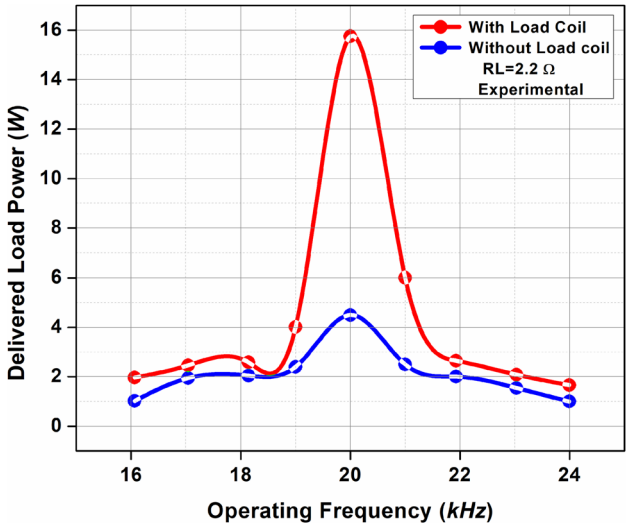


Fig. 7. Experimentally measured PDL in both two- and three-coil resonant WPT.

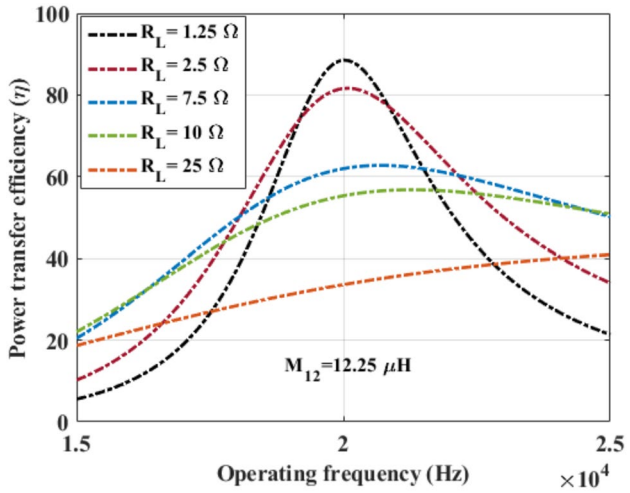


Fig. 8. PTE versus frequency in conventional WPT system.

couplings. In order to maintain PTE maximum, the ideal load resistance in both situations is extremely sensitive to the power transfer distance, dropping as the distance increases.As expected, power transfer efficiency increases with shorter transfer lengths in both cases. Nonetheless, the PTE in the two-coil system wins out under similar operating circumstances; this conclusion is further supported by the experimental findings displayed in Fig. 16.

The improvement in flexibility and controllability over power flow in load coil embedded three-coil structure through quick adjustment of the position of receiver coil and load coil with their impact on the PDL and overall system's PTE are elucidated in this discussion. Addition of an extra coil to the system in fact improve the system's flexibility by facilitating user the freedom of coil's placement adjustment which is usually mapped to the mutual coupling of the system. In addition, the power distribution among the loads and the system efficacy are observed to be very much sensitive with respect to the mutual inductances between the coils. Moreover, the criteria of optimal mutual coupling between transmitting coil, receiving coil and the load coil with their inclusive correlation correspondence to PDL and PTE are depicted in Fig. 17 (a-c). The optimal mutual inductance between receiver and load coil in load coil embedded three-coil WPT system will be calculated as:

$$\begin{cases} \dfrac{\partial \eta}{\partial M_{rl}} = 0 \\ M_{rl_Optimal} = \dfrac{1}{\omega}\sqrt{R_L\left(\sqrt{R_0^2 + (\omega M_{tr})^2}\right)} \end{cases} \tag{29}$$

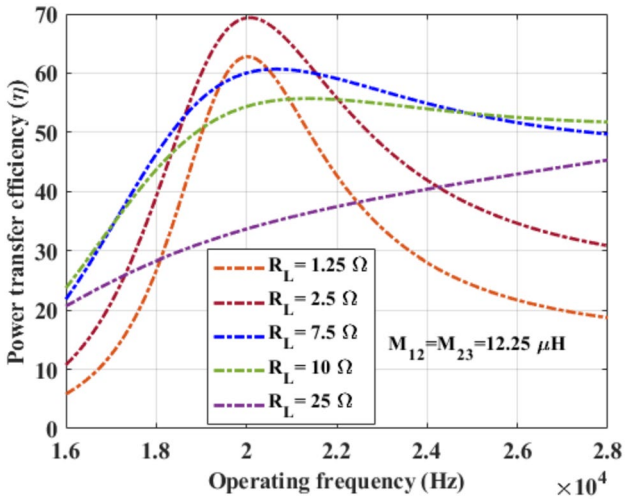


Fig. 9. PTE versus frequency in load coil involved (three-coil) WPT system.

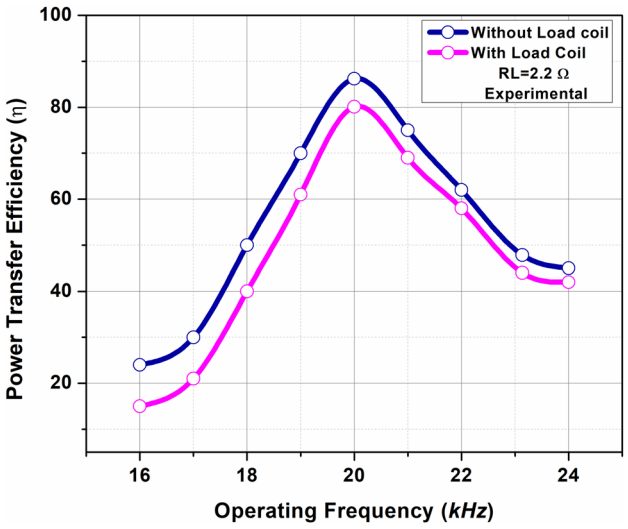


Fig. 10. Experimentally measured PTE in both two- and three-coil resonant WPT.

In order to describe the load characteristics of delivered load power, the model results are taken into consideration for three distinct sets of mutual coupling between receiver and load coil (M_{23} = *5, 12.5 & 17.5 μH*) data at resonant frequency (f_r = *20.2 kHz*) with varying mutual coupling between transmitting and receiving coil (M_{12}). The results show that the maximum delivered load power under tuned conditions is determined by a particular combination of factors such as R_L, M_{12}, and M_{23}. Additionally, a greater driving load necessitates a bigger air gap between the transmitter and receiver coils (smaller M_{12}) while the receiver coil and load coil are fixed (M_{23} = constant), and vice versa.

The clear association between mutual coupling and optimal load resistance is revealed by comparing the findings as displayed in Fig. 17(a–c). This is because the ideal load value varies depending on the set of parameters and also rises as the wireless link distance grows. Furthermore, whereas a bigger electric load is necessary for short-range wireless power transfer distances, a smaller electric load is required to supply power over a comparatively greater distance.

Following the description above, Fig. 18 (a-c) shows the load characteristics of PTE for load coil embedded three coil structures under comparable operating conditions. The graphic illustrates how the ideal PTE is determined in each scenario using a particular set of operating parameters. As is typical, under resonance conditions, the PTE increases as the air gap between the coils decreases. As the air-gap (or mutual coupling) between the coils narrows, the optimal load value progressively drops.

The experimental findings for different M_{12}&M_{23} are shown in Figs. 19 and 20 to support the load characteristics of PTE and PDL as previously mentioned. The results clearly show the ideal load situation, which changes depending on where the transmitting, receiving, and load coils are located. The ideal load values were found to change toward higher values with greater values of M_{23}, while maintaining a fixed physical distance of

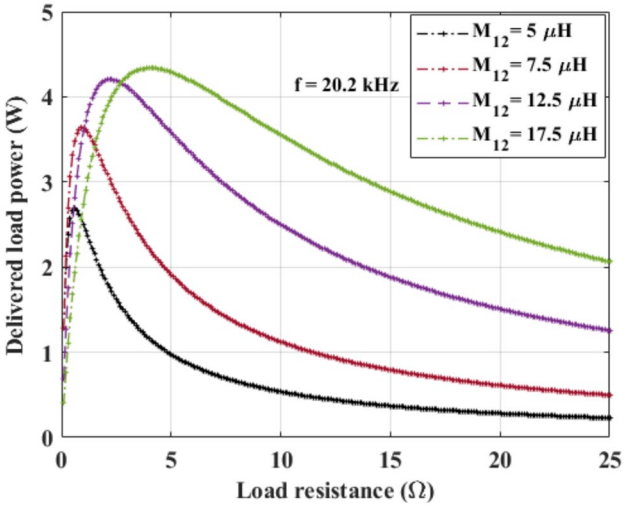


Fig. 11. PDL versus electric load corresponding to various mutual coupling in conventional WPT system at resonance frequency.

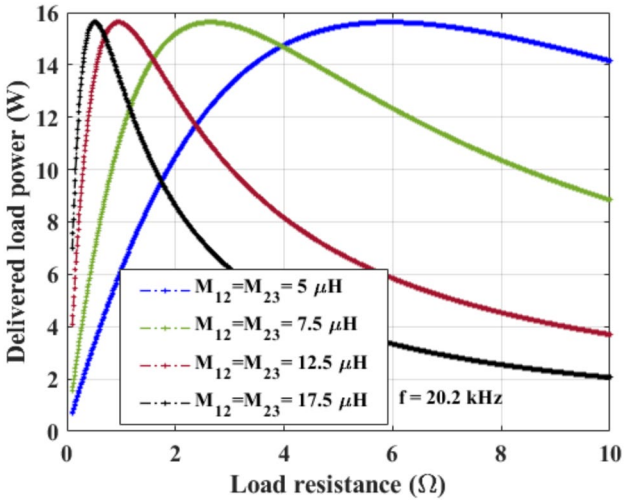


Fig. 12. PDL versus electric load corresponding to various mutual coupling in load coil involved (three-coil) WPT system at resonance frequency.

the wireless link. For instance, at an operating frequency of *20.2 kHz*, the system attains a maximum efficiency of 85% and 72% for mutual inductance M_{23} = *12.25 uH* and *5.2 uH*, respectively, with load values of *0.86 Ω* and *1.2 Ω*. Likewise, at ideal load values of *0.2 Ω* and *0.4 Ω*, respectively, the maximum PDL under comparable operating conditions is found to be *15 W*. Therefore, it is determined that the maximum PTE and PDL in load coil embedded three coil constructions are greatly influenced by the wireless link distance in relation to the load value.

The results supporting validation at higher frequency are depicted in Fig. 21 (a-b). The results show, the delivered load power is significant for lower frequency range at smaller load value. However, at higher frequency range, frequency splitting phenomena occurs and very negligible power is transferred to the load. Hence, it can be concluded that the suggested method is well appropriate for transferring power wirelessly at low frequency and the range of frequency is defined by the resonant frequency of the transmitting and receiving coil of the wireless link.The prime objective of this research was to resolve the major challenges of conventional two-coil WPT such as limited power transfer distance and inadequate control over power flow to the load through the proposed load coil embedded 3-coil WPT system. The coil design specification (Litz wire, 36 AWG) is considered for the proposed prototype so as operate within the frequency range below (< 50 kHz). The simulation results well validated through the experimental findings at lower frequency range (< 25 kHz), which confirms the claim.

The aforementioned results obtained from the load coil embedded 3-coil WPT system well agree to overcome the foremost challenges of conventional two-coil WPT system. As per the acquired results, the competitive difference between 2-coil and load-coil based 3-coil WPT systems have been illustrated in Table 2.

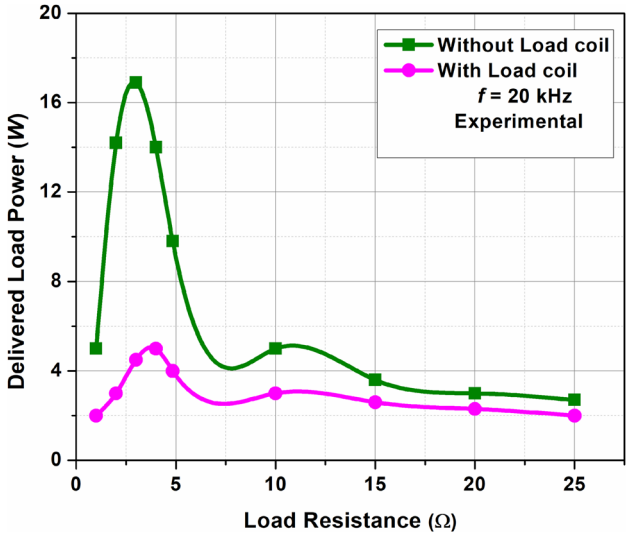


Fig. 13. PDL versus electric load corresponding to various mutual coupling in load coil involved (three-coil) WPT system at resonance frequency (Experimental).

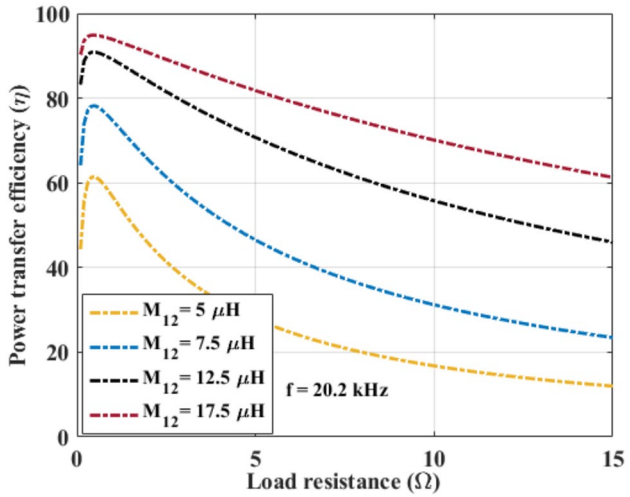


Fig. 14. PTE versus electric load & mutual coupling in conventional WPT system.

Conclusions

The present manuscript brings together mathematical modelling, analysis by utilizing circuit theory, experimental measurements and design insights of two-coil and load coil embedded 3-coil and multi-coil WPT systemswithin a single coherent framework. The notable features of three-coil configurationlie in demonstrating exceptional control over load power delivery capability, adaptability through swift mutual coupling adjustment, extended power transfer range with enhanced delivered load power without transmitter–receiver reconfiguration, supported by a fair analytical and experimental comparison with conventional two-coil WPT systemunder similar operating conditions. For instance, under identical operating conditions (resonant frequency of 20.2 kHz and load of *2.2 Ω)*, the suggested three-coil WPT system delivers an optimum load power of *15.82 W* with PTE 80% in contrast to *4.22 W* with PTE 86.4% obtained using the conventional approach. These features put the three-coil system in a strong position to compete not only in the field of traditional WPT method but also the incorporation of dedicated load coils offers a compact and scalable alternative for multi-load WPT system. Moreover, the implementation of three-coil structure may increase the applications of WPT technology, providing fresh chances for creativity and enhancement.

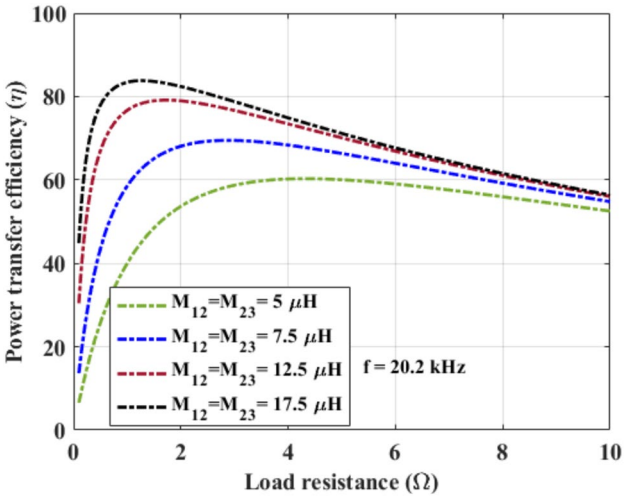


Fig. 15. PTE versus electric load & mutual coupling in load coil involved (three-coil) WPT system.

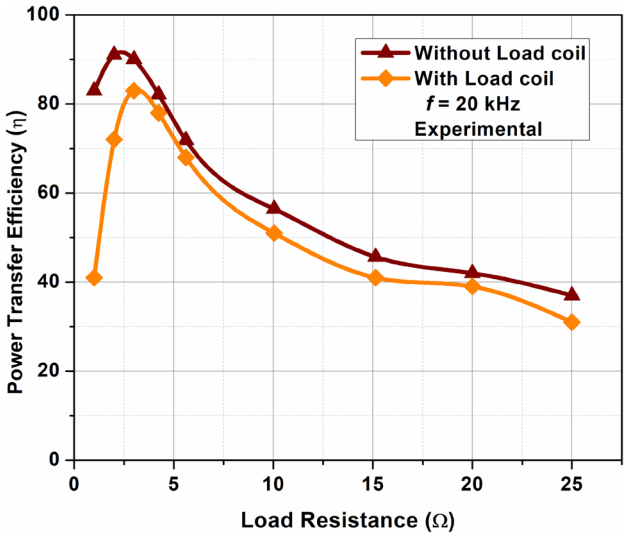
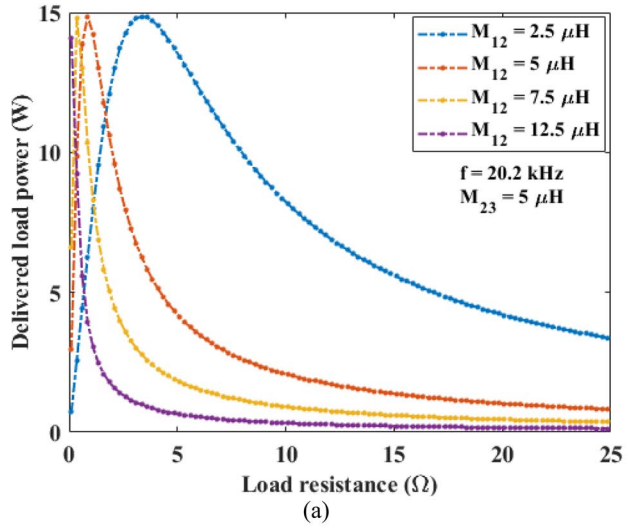
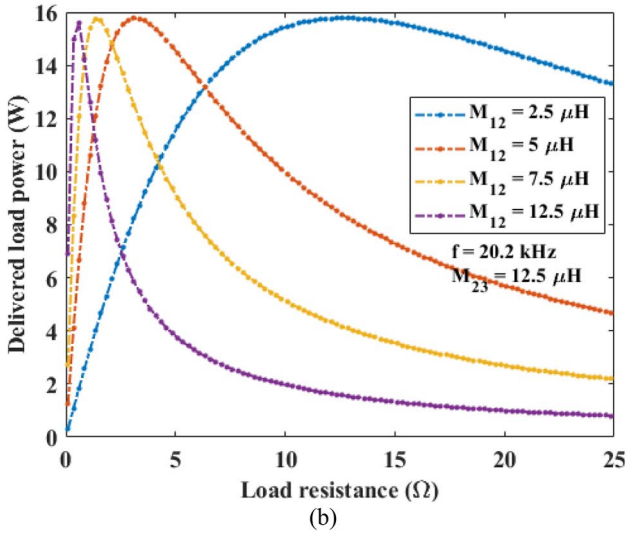


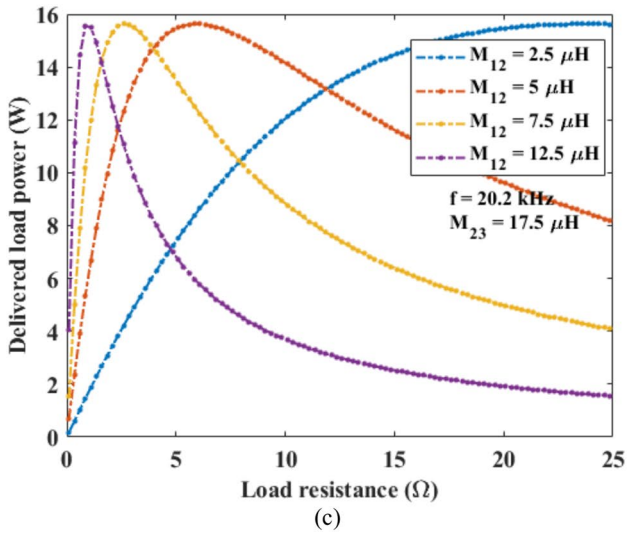
Fig. 16. PTE versus electric load in conventional WPT and load coil involved (three-coil) WPT system (Experimental).



(a)

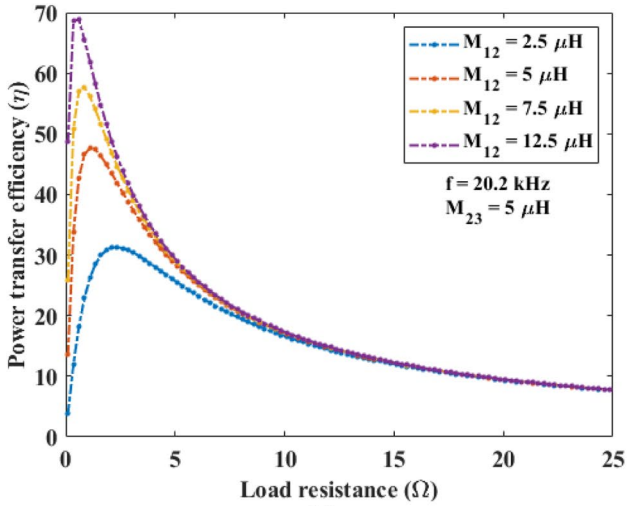


(b)

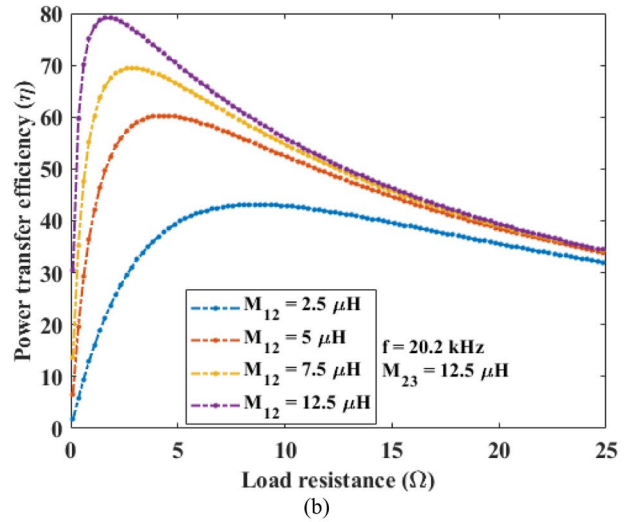


(c)

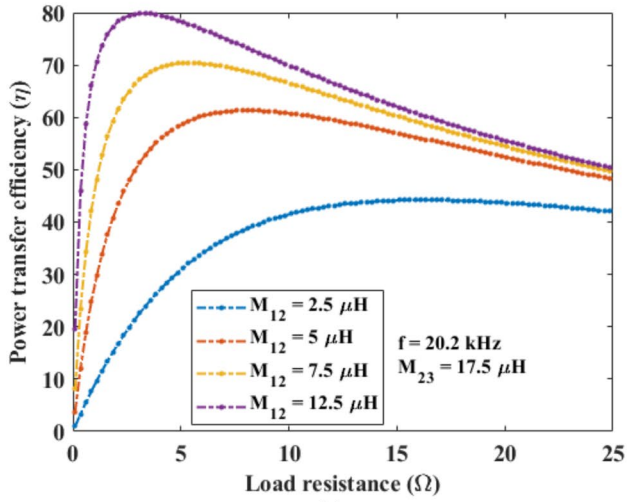
Fig. 17. (**a**), (**b**) & (**c**). Load characteristics of PDL with respect to varied coils' position.



(a)



(b)



(c)

Fig. 18. (**a**), (**b**) & (**c**). Load characteristics of PTE with respect to varied coils' position.

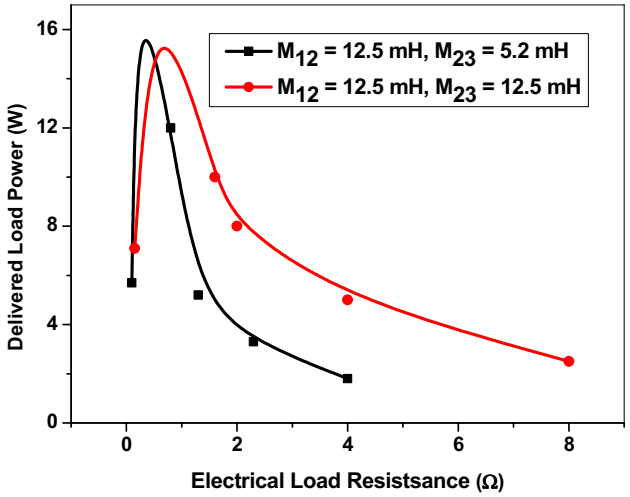


Fig. 19. Experimentally measured load characteristics of PDL.

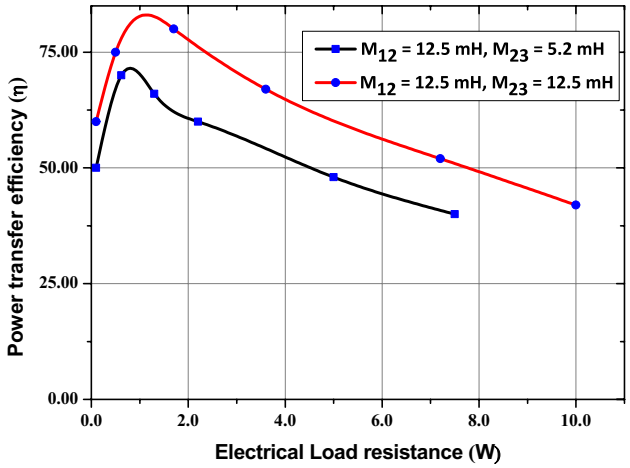


Fig. 20. Experimentally measured load characteristics of PTE.

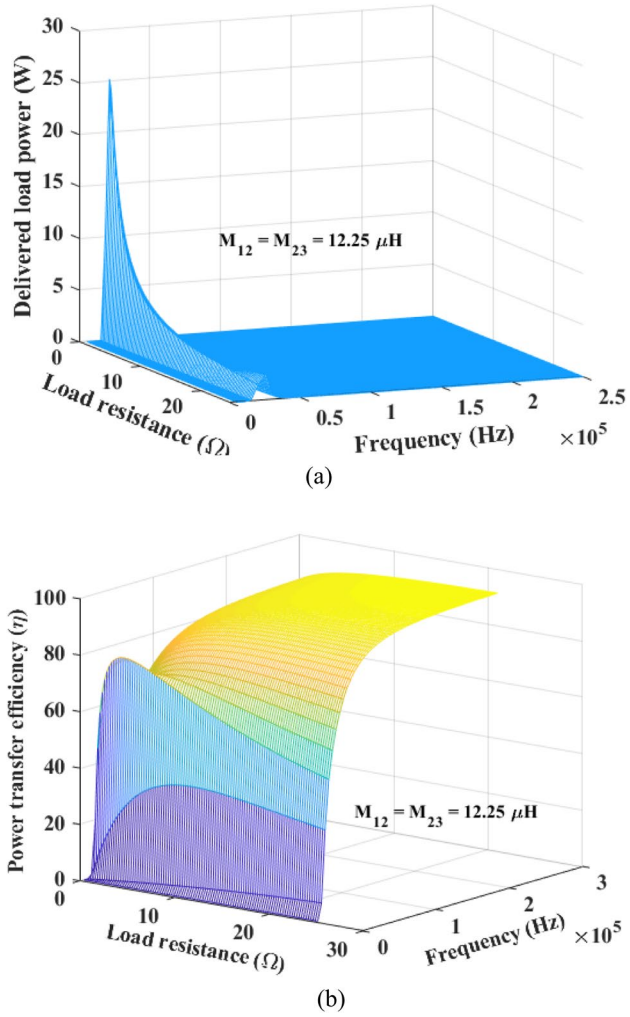


Fig. 21. (**a**-**b**): High frequency characteristics of delivered load power and power transfer efficiency.

Parameter	Two-Coil WPT system	Three-coil WPT system
Basic configuration	Transmitter–Receiver (Tx–Rx)	Includes an extra load coil Transmitter–Receiver-Load coil (Tx–Rx-Load)
Wireless power transfer distance	Short range	Extended power transfer range
Delivered load power	Decreases with distance (R_L = 2.2 Ω, M_{12} = 12.25 μH, P_L = ~ 4.22 W)	Comparatively higher (R_L = 2.2 Ω, $M_{12} = M_{23}$ = 12.25 μH, P_L = ~15.82 W)
Power transfer efficiency	Relatively higher (R_L = 2.2 Ω, M_{12} = 12.25 μH) (η = 86.4%)	Slightly lower (R_L = 2.2 Ω, M_{12} = 12.25 μH) (η = 80%)
Controllability	Reconfiguring/ redesigning Tx/Rx coils	Easy adjustment of Rx and load coil
Sensitivity to load variation	Highly sensitive	Better Load Tolerance
Sensitivity to frequency variation	Highly sensitive	Broadens the effective operating frequency region (frequency splitting)
Multi-load capability	Require multi-receiver coils	Supports scalable multi-load architectures
System complexity	Simple structure	Slightly complex due to additional load coil but improves functionality

Table 2. Competitive Difference between 2-coil and load-coil based 3-coil WPT systems.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 13 November 2025; Accepted: 3 April 2026

Published online: 29 April 2026

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Author contributions

We the authors, Pragyan P Mohanty, Sushree S Biswal, Durga P Kar and Satyanarayan Bhuyan declare that the article is original. Pragyan P Mohanty contributed to the conceptualization, system design, simulation, and data analysis. Sushree S Biswal developed the methodology, performed circuit modeling, and prepared the original manuscript draft. Durga P Kar conducted the experimental validation, manuscript visualizations and interpreted the results. Satyanarayan Bhuyan supervised the work, provided technical guidance, and performed manuscript review and editing. All authors have read and approved the final manuscript.

Funding

Open access funding provided by Siksha 'O' Anusandhan (Deemed To Be University).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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